

Technical Report: Rescue Mission Pipeline Optimization

1. Introduction

The objective of this project is to solve the "Victim Rescue Ordering" problem using a hybrid approach that combines traditional search algorithms with evolutionary computing. The challenge involves a robot that must rescue five victims on a 10×10 grid containing obstacles (walls). Since movement is constrained by these walls, the system must calculate real-world travel distances and then optimize the sequence of visits to minimize the total path length.

2. Algorithm Design (The Pipeline)

A. Step 1: Search Strategy (BFS)

The environment is represented as a 2D grid. To handle obstacles, we implemented the **Breadth-First Search (BFS)** algorithm as the first stage of our pipeline.

- **The Logic:** BFS explores the grid layer by layer to find the shortest possible path between any two points while avoiding cells marked as walls.
- **Distance Matrix:** We executed BFS for every possible pair (Robot to Victims, and Victims to each other) to build a **6×6 Distance Matrix**. This matrix serves as the "knowledge base" for the Genetic Algorithm, ensuring it works with true traversal costs rather than simple coordinates.

B. Step 2: Genetic Algorithm (GA) Optimization

The GA is responsible for the optimization part of the pipeline—finding the best permutation of victims.

- **Chromosome Representation:** We used **Permutation Encoding**. Each chromosome is a list of five unique integers (e.g., [3, 1, 5, 2, 4]), where each

number represents a victim's ID. This ensures the robot visits every victim exactly once.

- **Population:** A set of 20 random permutations was generated as the initial generation.

3. Fitness Function (Evaluation Logic)

The fitness function is the core of our evaluation. Since our goal is to minimize the total distance, a "better" solution is one with a lower total cost.

Mathematical Logic:

The total fitness of a given chromosome (permutation) is calculated as:

$$TotalDistance = Dist(R, V_{P1}) + \sum_{i=1}^4 Dist(V_{Pi}, V_{P(i+1)})$$

- R : The initial starting position of the Robot. 
- V_{Pi} : The i -th victim in the chromosome's sequence. 
- $Dist$: The shortest path distance retrieved from the **BFS Distance Matrix**. 

Reward and Penalty Mechanism:

- **Rewarding:** The algorithm rewards chromosomes that sequence victims in a way that minimizes travel steps. These solutions are prioritized for selection.
- **Penalizing:** Chromosomes with inefficient paths result in a high "Total Distance" value and are naturally filtered out through the selection process.

4. GA Parameters & Operators

- **Population Size:** 20.
- **Generations:** 100.
- **Selection:** Rank-based selection.
- **Crossover (Ordered Crossover):** Due to the permutation nature, we implemented **Ordered Crossover** to maintain valid sequences and avoid duplicate victims while combining parental traits.
- **Optimization Technique (Mutation):**

- **Type: Inorder Mutation (Swap Mutation).**
- **Logic:** We implemented **Swap Mutation** (interchanging two random positions). This is the optimal choice for permutations as it maintains the integrity of the sequence (no duplicates) while introducing diversity to escape local optima.

5. Comparison & Results

- **Comparison:** The BFS algorithm was essential for **local optimality** (shortest path between two points), while the GA provided **global optimality** (best overall order). The GA found the solution significantly faster than an exhaustive manual search.
- **Results:** The pipeline successfully converged to an optimal sequence with a total travel cost of **45.0 steps**.

6. Visualization

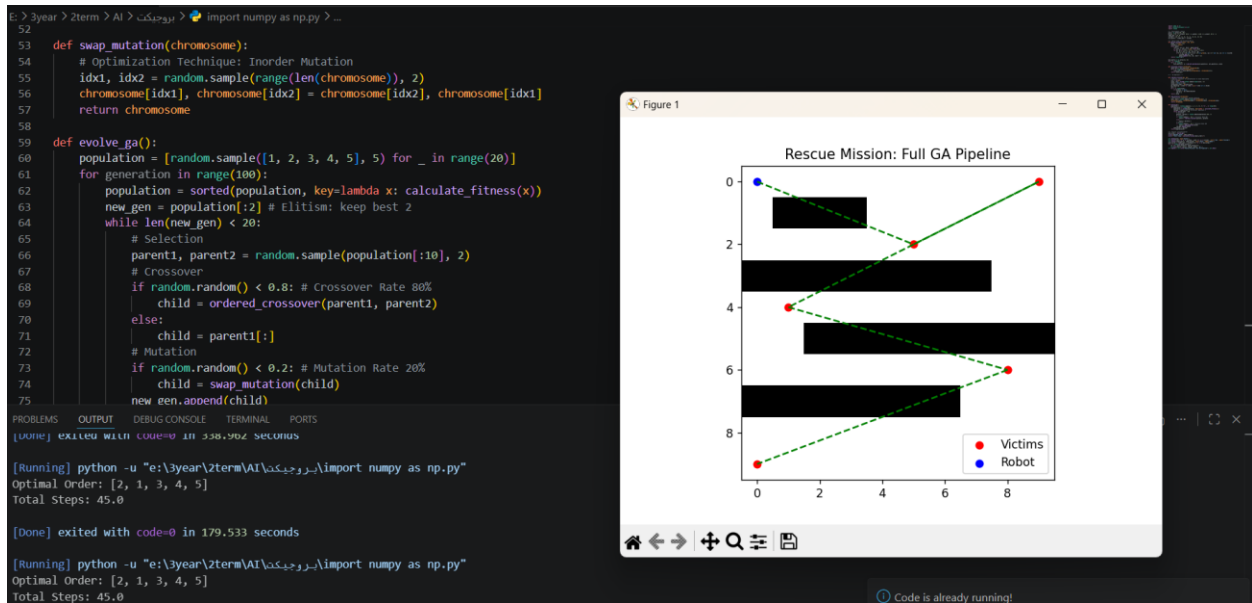


Figure 1: Optimized Rescue Path. The dashed green line indicates the optimized sequence found by the Full GA Pipeline, successfully navigating around all walls.

8. Conclusion

By combining BFS for local pathfinding and GA for global sequence optimization, we successfully solved the rescue mission problem. This hybrid pipeline proves that evolutionary algorithms can significantly enhance decision-making in complex, constrained environments.